



Molecular Biology

“Gene and Genome of Human”

SGL - 1

Hawler Medical University
College of Medicine

Dr. Bejan Ahmed DIZAYEE
M.Sc. Medical Microbiology / Molecular Biology
Director of Central Veterinary Laboratory in Kurdistan Region

Molecular Biology :



Domain of life

A. Eukaryotic cells.

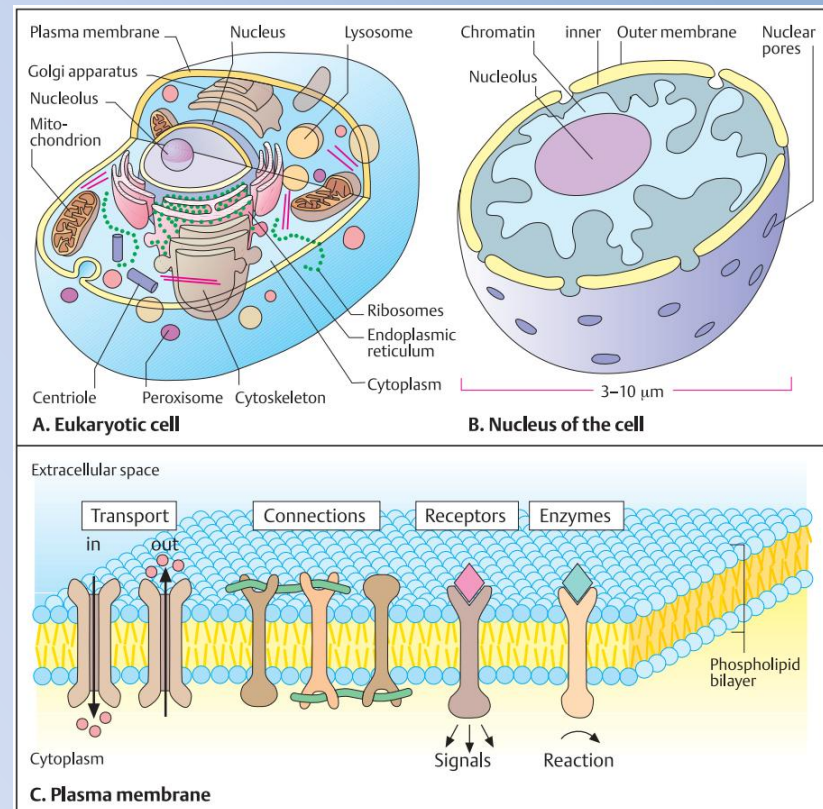
B. Prokaryotic Cells.

C. Archaea

Molecular Basis of Genetics

A. Eukaryotic cells:

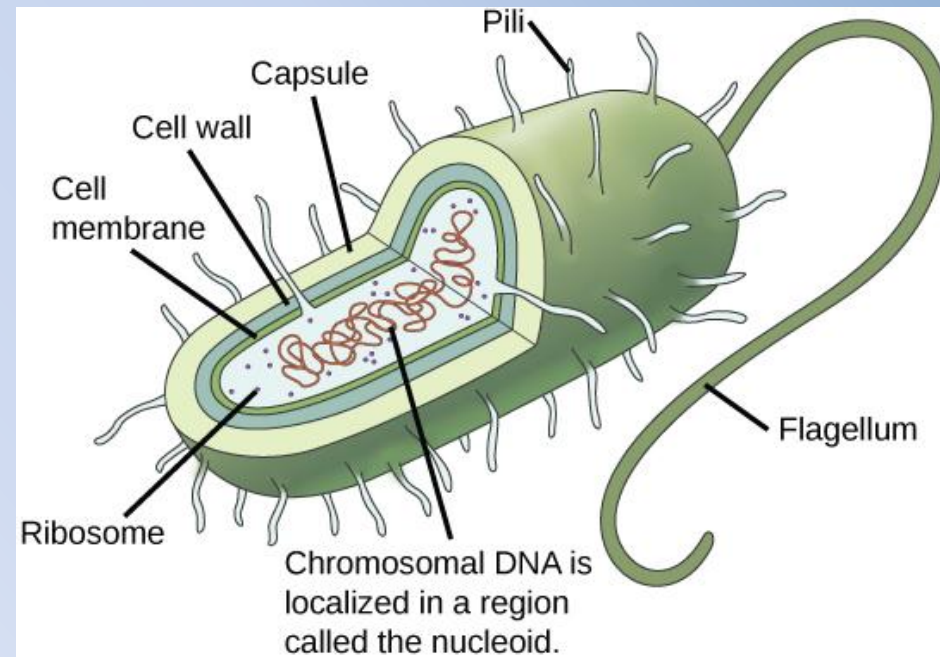
- Found in animal cells, plants, fungi and protists
- Cell with true nucleus, genetic material surrounded by a membrane
- Eukaryotic genome is more complex than that of Prokaryotes and distributed among multiple chromosomes
- Eukaryotic DNA is linear
- Eukaryotic DNA is complexed with proteins called histones
- Numerous membrane-bound organelles
- Complex internal structure
- Cell division by mitosis



Molecular Basis of Genetics

B. Prokaryotic cells

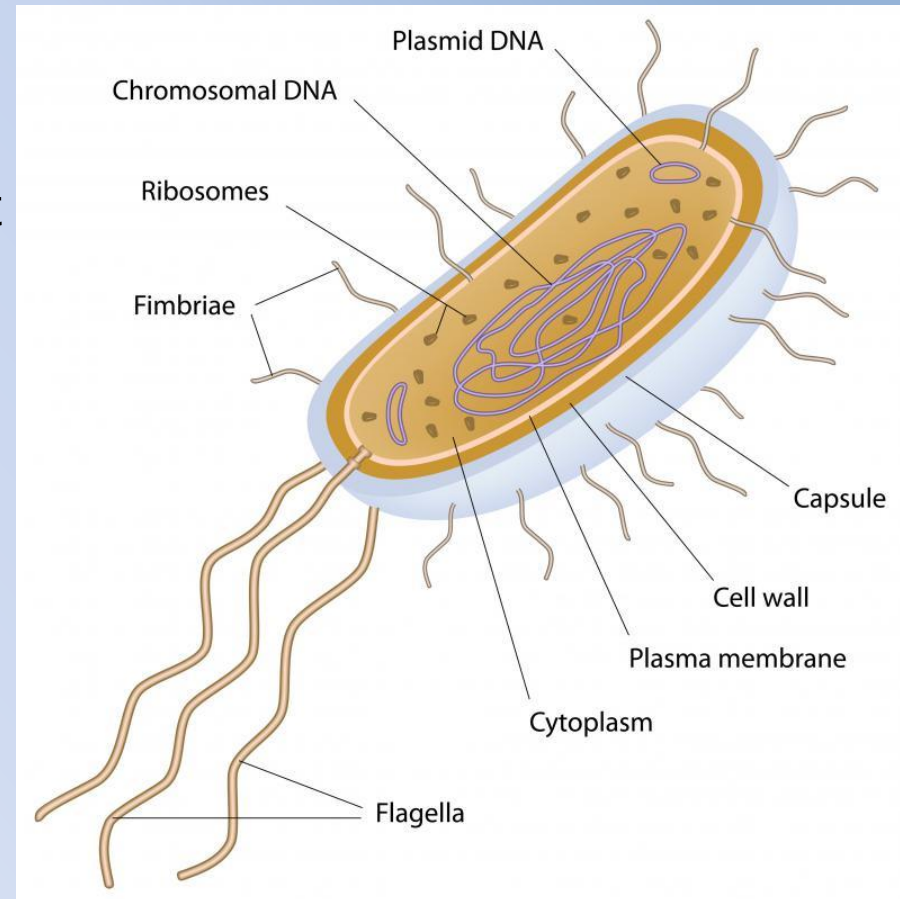
- Unicellular organisms, found in all environments.
- Without a Nucleus, no nuclear membrane (genetic material dispersed throughout cytoplasm)
- No membrane bound organelles
- Cell contain only one circular DNA molecule
- DNA is naked (No Histon)



Molecular Basis of Genetics

C. Archaea

Archaea is prokaryotes organisms but some aspects of molecular biology more similar those of Eukaryotes





Molecular biology

DNA _Carrier of Genetic Information

1869: Friedrich Miescher:

A new, acidic, phosphorus containing substance made up of very large molecules named “nuclein”, its biological role was not recognized.

1889: Richard Altmann:

Introduced the term “nucleic acid”

1900:

The purine and pyrimidine bases were known.

1920:

The two kinds of nucleic acids, RNA and DNA, were distinguished.

1944:

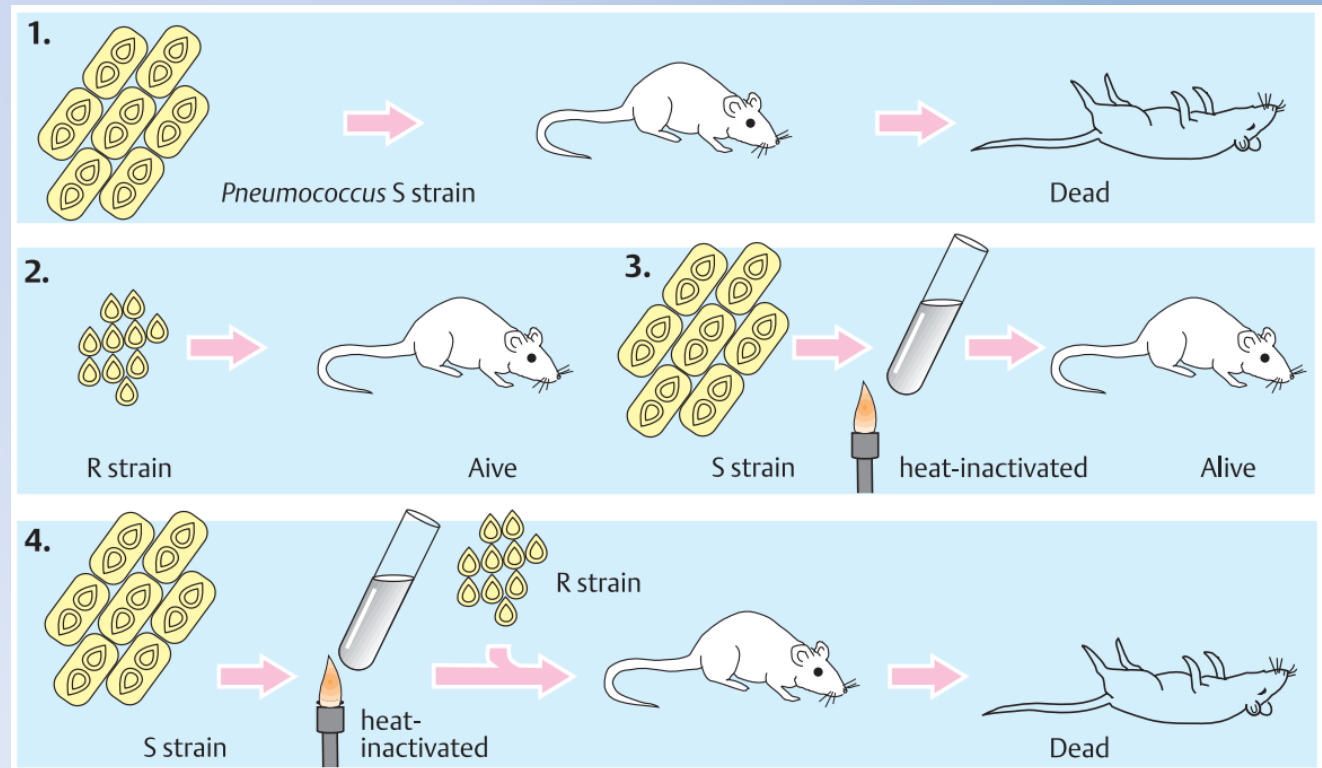
Indicated that DNA could be the carrier of genetic information.

A. The observation of Griffith (1928)

Indicated that DNA could be the carrier of genetic information.

In 1928 the English microbiologist Fred Griffith made a remarkable observation through investigating various strains of:

Pneumococcus
strain S (smooth)
strain R (rough)



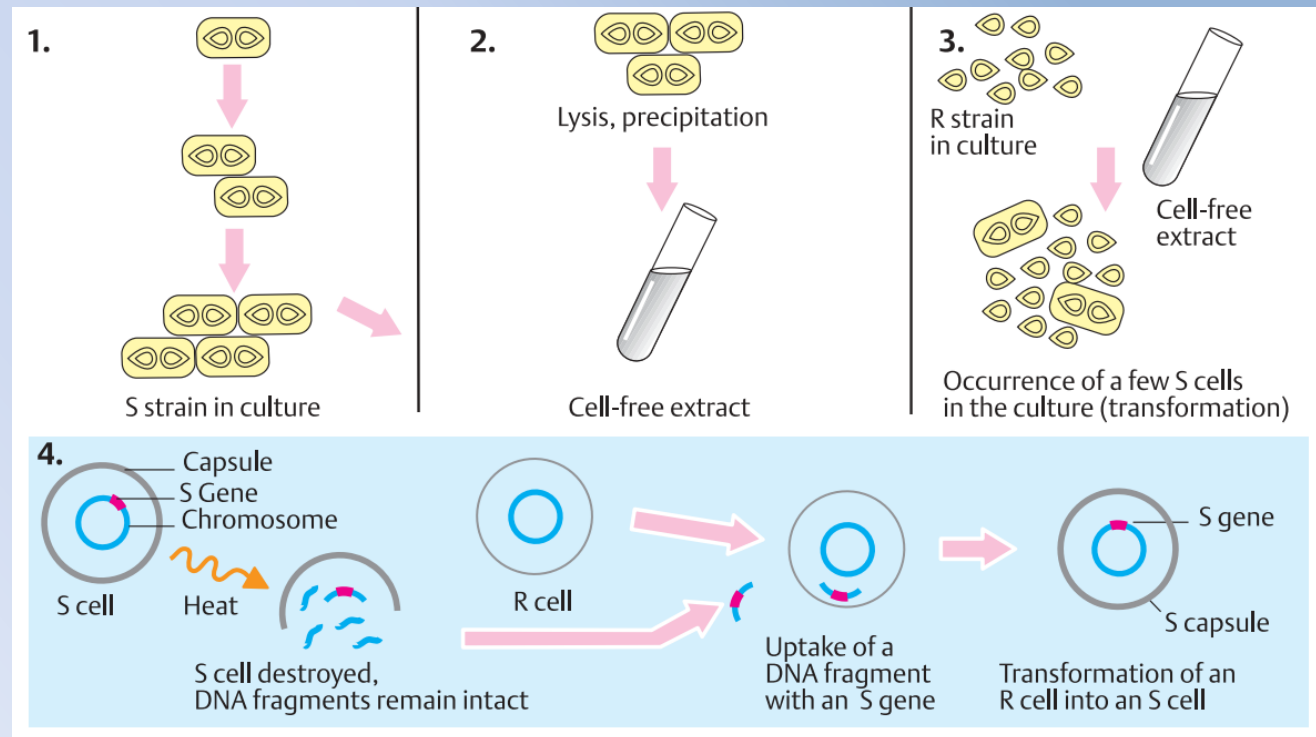
Apparently, cells of the *R strain* were changed into cells of the *S strain*

Transformed

B. The transforming principle is DNA

Griffith's findings formed the basis for investigations by Avery, MacLeod, and McCarty (1944); explained the chemical basis of the transforming principle

Pneumococcus
strain S (smooth)
strain R (rough)



Avery and co-workers determined that this was attributed to the DNA alone. Thus,

the DNA must contain the corresponding genetic information

This explained Griffith's observation

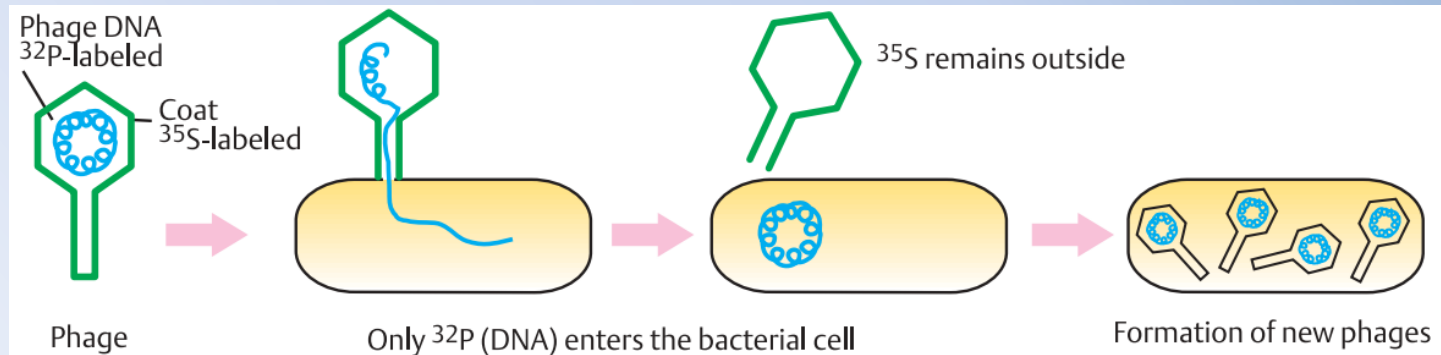
C. Genetic information is transmitted by DNA alone

The final evidence that DNA, and no other molecule, transmits genetic information was provided by Hershey and Chase in 1952.

They labeled:

Bacteriophages capsular protein of with radioactive sulfur (^{35}S)

Bacteriophages DNA with radioactive phosphorus (^{32}P).



Complete phage particles in the cell proved that

DNA was the exclusive carrier of the genetic information

needed to form new phage particles, including their capsular protein

Overview of Organizations of Life

Nucleus = library

Chromosomes = bookshelves

Genes = books

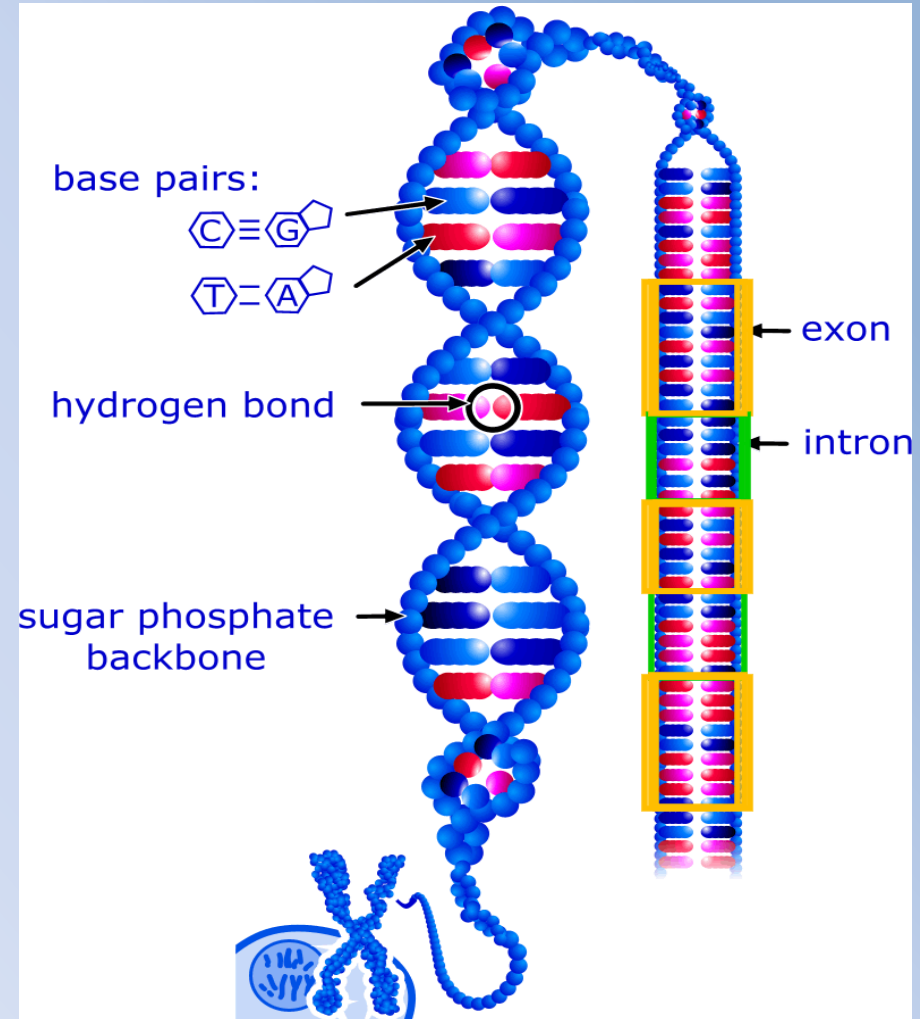
- ❖ Almost every cell in an organism contains the same libraries and the same sets of books.
- ❖ Books represent all the information (DNA) that every cell in the body needs so it can grow and carry out its various functions

~3.2 billion base pairs in every cell build the human genome

Genes form only 1.5% of the human genome

A gene is a segment of the DNA, that encodes the construction plan for a protein

In humans there are 30,000 genes only



Molecular biology is:

- The study of gene structure and function of organisms at the molecular level
- The study of molecular basic of the process of **Replication**, **Transcription** and **Translation** of the genetic material.
- The four main families of biochemical molecules in cells :

Carbohydrates
Proteins

Lipid (fats and oils)
Nucleic acid (DNA and RAN)

Nucleic Acids and Nucleotides:

- Nucleic acids are macromolecules (DNA and RNA)
- Central to the storage and transmission of genetic information.
- Nucleotides are the subunits (monomers) of DNA and RNA.
- Essential for all known forms of life (reproduction)

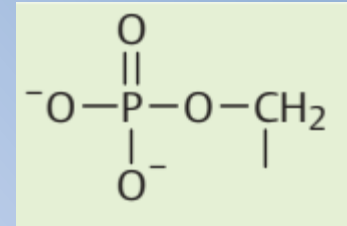
Functions of nucleic acid:

- Convey energy (e.g. ATP, GTP),
- Encoding, transmitting and expressing genetic information
- They are part of essential coenzymes.
- They regulate numerous metabolic functions (enzymes, RNA).

Molecular Biology :

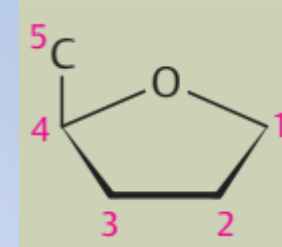
Nucleotides are composed of:

1- Phosphate group.



2- Pentose sugar:

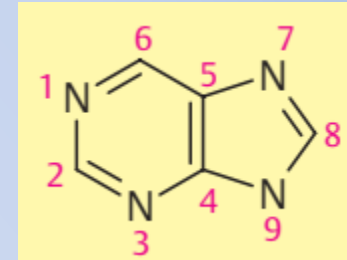
- a. Ribose (RNA)
- b. Deoxyribose (DNA)



3- Nitrogen containing bases

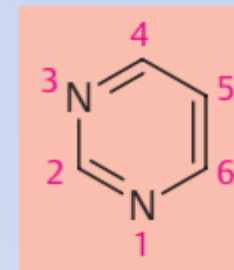
Purine- two rings

- Adenine (A)
- Guanine (G)



Pyrimidine – contains one ring

- Cytosine (C)
- Thymine (T)
- Uracil (U)



DNA: A,G,C,T

RNA: A,G,C,U